

Haomiao Li

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EDUCATION

Yale University, New Haven, Connecticut

2016 - 2017

Masters in Statistics

The College of William and Mary, Williamsburg, Virginia

2012 - 2016

B.S. in Mathematics; B.S. in Finance

WORK EXPERIENCE

Pinterest, Palo Alto, CA

05/2022 - present

Tech Lead, Related-Pins Recommendation

- Developed and productionized many model iterations on the closeup ranking and blending models e2e into production, including transformer-based ranking model, multi-gate mixture of experts module, user sequence signals encoding with transformer, model-based personalized utility ([eng blog](#)), large ID embedding tables which cumulatively brought 10%+ relative engagement gains
- Developed latest modeling techniques in closeup ranking model such as DHEN, model scaling, DCN v2, etc
- Collaborated with the infra team to apply various optimizations to reduce model training time and online serving latency.
- Tech lead the Ranking & Blending team for closeup surface (the biggest organic surface, serves >50% organic traffic) on roadmap drafting, OKR planning and tracking, resolving blockers to drive impact and ensure deliveries

Machine Learning Engineer, Ads Ranking

11/2020 - 05/2022

- Developed video ads prediction models from GBDT to DNN+AutoML, improving video view CTR metrics with an \$1M+ annual infra cost savings by optimized ETLs
- Collaborated with ads infra team to launch distributed training functionality for L2 model training, which halved model training time and solved the long-standing pain point
- Launched transformer-based ads engagement model on home feed and related pins surface into production

Uber Technologies., Inc, San Francisco, CA

01/2019 - 11/2020

Applied Scientist, Machine Learning, Uber Maps

- Developed transformer-based NLP Models for joint intention detection and slot fillings on messages between rider and driver to detect potential pickup location issues and names, and helped the POI index expansion;
- Built the universal pickup suggestion ML model at Uber Maps, which empowered the pickup for non-Uber Pool Rides globally. The system suggested the best point among candidates for riders to take the ride given the raw device location and its surrounding geographic, tried to provide the frictionless experience
- Responsible for the model end-to-end, from offline feature engineering and feature computation to deploy features into the real time system, including real-time computation and dispersal
- Designed numerous experimentations and A/B test analysis to evaluate the ML model's relevant online metrics, as well as other relevant product changes experimentations

BMW North America, Mountain View, CA

07/2017 - 01/2019

Machine Learning Engineer

- Built real-time AWS platform to provide API access to support customer dashboards, and fast queryable databases
- ETA prediction using Xgboost, and reduced the mean absolute error by 40 seconds.

PUBLICATIONS

Effects of harvesting quota and protection zone in a reaction-diffusion model

2014

- Constructed algorithms in Matlab to perform numerical simulations on PDE systems for fish population
 - **[Publication]** Cui R., Li H., Mei L. and Shi J. "Effect of harvesting quota and protection zone in a reaction-diffusion model arising from fishery management," *Discrete and Continuous Dynamical Systems: Series B*, Vol. 22, no. 3, 2017, pp. 791 - 807.

SKILLS

- **Machine Learning:** Deep Learning, Statistics, NLP, Recommendations and Rankings, Forecasting and Regressions
- **Programming:** Python, Spark, Scala, Pytorch, SQL